

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I D.G.Raghavi (192521173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.G.Raghavi (192521173)

Assessment Tasks Short Answer Test

- 1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.**

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet.

Four characteristics:

1. On-demand self-service: Users can access computing resources whenever needed without contacting the service provider.
2. Broad network access: Cloud services can be accessed through the internet using computers, tablets, and mobile phones.
3. Resource pooling: Computing resources are shared among multiple users, improving resource utilization.
4. Rapid elasticity: Resources can be increased or decreased quickly based on user demand.

Difference: Traditional on-premise computing requires organizations to purchase and maintain their own hardware, whereas cloud computing provides resources over the internet.

2. **Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.**

Cloud computing developed through several stages:

1. Mainframe computing: Large central computers provided computing power to multiple users.
2. Personal computers: Computers became smaller and more affordable, allowing individuals to perform their own computing tasks.
3. Client-server computing: Clients requested services from centralized servers through networks.
4. Virtualization: A single physical server could run multiple virtual machines, improving hardware utilization.
5. Internet evolution: Faster internet connections enabled users to access services and data remotely.
6. Web-based software: Applications became accessible through web browsers without requiring complete local installation.
7. Modern cloud computing: Providers such as AWS, Microsoft Azure, and Google Cloud began offering computing resources and applications on demand.

Conclusion: Advances in hardware, virtualization, networking, and internet-based software enabled scalable, flexible, and cost-effective cloud platforms.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization is a technology that divides one physical server into multiple virtual machines. Each virtual machine can run its own operating system and applications.

Role of server virtualization:

- Improves hardware utilization.
- Reduces hardware and maintenance costs.
- Allows multiple operating systems to run on one server.
- Enables flexible resource allocation.
- Supports rapid creation and management of virtual machines.

Difference:

Virtualization	Cloud Computing
A technology that creates virtual resources.	A service model that delivers computing resources over the internet.
Can operate within a local data center.	Commonly provides remote, on-demand services.
Focuses on resource abstraction and sharing.	Focuses on service delivery, scalability, and metered usage.
Example: VMware ESXi.	Example: Amazon Web Services (AWS).

Conclusion: Virtualization is an important enabling technology for cloud computing, while cloud computing uses technologies such as virtualization to deliver services to users.

4. Discuss the role of data centers in delivering cloud services..

A data center is a facility that contains servers, storage systems, networking equipment, and other infrastructure used to store, process, and manage data.

Role of data centers:

1. Data storage: Stores large amounts of user data and application information.
2. Processing power: Provides servers to run applications and process user requests.
3. Networking: Connects servers and enables users to access cloud services through the internet.
4. Security: Uses access controls, monitoring, and security systems to protect data.
5. Availability: Uses backup power, redundant equipment, and disaster recovery systems to reduce service interruptions.
6. Scalability: Allows cloud providers to add resources as user demand increases.

Example: When users store files in Google Drive, data centers provide the storage, processing, and networking infrastructure needed to access those files.

Conclusion: Data centers form the physical foundation of cloud computing by providing the infrastructure required for reliable, secure, and scalable cloud services.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Model	Full form	Description	Example
IaaS	Infrastructure as a Service	Provides virtual servers, storage, and networking.	Amazon EC2
PaaS	Platform as a Service	Provides a platform to develop, test, and deploy applications without managing the underlying infrastructure.	Google App Engine
SaaS	Software as a Service	Provides ready-to-use software over the internet.	Gmail

XaaS	Everything as a Service	Provides different IT resources and capabilities as services over a network.	Microsoft 365
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Key differences:

- IaaS: Users manage the operating system and applications, while the provider manages the physical infrastructure.
- PaaS: Users focus on application development, while the provider manages the platform and infrastructure.
- SaaS: Users access and use the software; the provider manages the application and its infrastructure.
- XaaS: A broad term covering many services delivered as needed, including storage, security, networking, and software.

Conclusion: IaaS provides infrastructure, PaaS provides a development platform, SaaS provides ready-to-use software, and XaaS represents the broader concept of delivering IT capabilities as services.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow